NAKAYAMA IN THE GRADED SETTING: GENERATION MODULO THE POSITIVE-DEGREE PART

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Homogeneous generators and their degrees. Used at the start of the counted proof (lines 13481-13483). Lets one generator of minimal degree be singled out, which is what forces it to vanish.
- Reduction of the general statements to the first one. Used in the second half of the counted proof (lines 13485-13488). Applies the first assertion to a quotient or to a chosen submodule to obtain parts (2), (3) and (4).

Lemma 0.1. *Let S be a graded ring. Let M be a graded S-module.*

(1) *If $S_+M = M$ and M is finite, then $M = 0$.*
(2) *If $N, N' \subset M$ are graded submodules, $M = N + S_+N'$, and N' is finite, then $M = N$.*
(3) *If $N \to M$ is a map of graded modules, $N/S_+N \to M/S_+M$ is surjective, and M is finite, then $N \to M$ is surjective.*
(4) *If $x_1, \ldots, x_n \in M$ are homogeneous and generate M/S_+M and M is finite, then $x_1, \ldots, x_n$ generate M.*

Proof. Proof of (1). Choose generators $y_1, \ldots, y_r$ of M over S. We may assume that y_i is homogeneous of degree d_i. After renumbering we may assume $d_r = \min(d_i)$. Then the condition that $S_+M = M$ implies $y_r = 0$. Hence $M = 0$ by induction on

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r. Part (2) follows by applying (1) to M/N. Part (3) follows by applying (2) to the submodules $\mathrm{Im}(N \to M)$ and M. Part (4) follows by applying (3) to the module map $\bigoplus S(-d_i) \to M$, $(s_1, \ldots, s_n) \mapsto \sum s_i x_i$. $\square$